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Adaptive switching and routing protocol design and optimization in internet of things based on probabilistic models

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ABSTRACT

Through smart cities, Intelligent Transportation Systems (ITS), the agricultural sector, and wearable devices, the Internet of Things (IoT) has revolutionized several human interests. Through the development of new cluster tasks, the Decision-Making System (DMS) of Cluster Heads (CHs), and improving the accuracy of traffic prediction and reliability of transportation, the present study intends to improve the energy depletion of IoT devices. The paper explores the subject of data flow optimization using Fuzzy Assisted Cuckoo Search Optimization (FACSO), traffic flow using Gaussian Process Regression (GPR), and CH prediction using the Stochastic Optimization Algorithm (SOA). Optimizing network lifetime while minimizing Energy Consumption (EC) is feasible through the practical application of the SOA, GPR, and FACSO models. Increasing End-to-End Delay (EED), Network Throughput (NT), and energy efficiency can be rendered feasible through a real-time DMS regarding routing employing a novel approach referred to as FACSO. This approach has enhanced the efficacy and reliability of Wireless Sensor Networks (WSN). With up to 500 nodes and an EC of 0.3451 *J*, the experiment's findings demonstrate that a proposed SOA-FACSO model achieves superior EED.

1. Introduction

The Internet of Things (IoT) has got attention in recent years. It is a potential technology that can provide remedies to challenges that arise in multiple sectors. The Internet serves as a network's communication backbone for exchanging information from one platform to another globally. The IoT concept came into existence in 1998, and in 1999, Kevin Ashton coined the term IoT at the Massachusetts Institute of Technology (United States) Auto-ID lab [1]. It's an impenetrable but intelligent system that detects, regulates, and can be reprogrammed through embedded systems to communicate with one another. With high productivity improvements, IoT allows instant access to information related to any device. According to the CISCO team, around 50 billion smart devices have already been connected to the Web [2].

A billion connected devices comprise the IoT, which can detect, collect, and send data from one device to another without human support. Health care tracking, building automation, logistics, linked autos, smart city development, smart grid, smart home, smart retail, smart agriculture, and other IoT-enabled services all contribute to ameliorating people's lives. The Internet will become obsolete in this scenario, and a new idea of connected smart' things will emerge. IoT can transform the Internet so that Machine-To-Machine (M2M) learning will become a reality. The reconfiguration will ensue by making physical devices' smart,' allowing them to accomplish things independently, resulting in the emergence of the IoT. By connecting anywhere at any time but in any location, IoT promises to make advanced devices more available.

IoT primarily allows for the autonomous yet safe communication of real-world objects. IoT reduces physical work by automating regular processes. The number of Internet-connected objects is continually expanding. Smartphones contain a range of sensors that detect data, compute it, and send the relevant data over the Internet. Using that system with diverse devices and sensors, several applications can be made to provide substantial benefits. IoT intelligent things can be uniquely identified. These gadgets use Radio-Frequency Identification (RFID) tags or barcodes, which are detected by sensor devices [3].

The data acquired by the sensors is sent to the processing system over the Internet for analysis. The processed findings are forwarded to the Decision-Making System (DMS) and Action-Invoking System (AIS), which performs the appropriate action. Important information and facilities are constantly available, making it easy to design new apps, introduce new work, connection, and performing methods, and contribute to a new lifestyle. As a result, there will be much traffic, and humans will be in the minority of traffic generators and recipients [4].

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Because of the issues and opportunities that IoT brings, it is being researched in numerous research areas. In this instance, maintaining a consistent model is required to store all the data efficiently.

Routing is the method of sending data from one node to another. The routing protocol's performance is accomplished by lowering the route cost. Efficient route selection saves energy during packet delivery and extends the Network's Lifetime (NL) [5]. Route discovery and maintenance protocols provide information about the journey from the source node (SrN) to the destination (DsN). The node is usually in one of two modes: storing or non-storing. The participant node collects and transmits data to its parent in the storage mode. In non-storing mode, the participant node conveys data to its parent node. Proactive, reactive, and hybrid protocols are route discovery and maintenance [6].

The routing procedure is accepted based on the node's location in a network. Network structure-based routing enhances the Quality of Service (QoS) while lowering network node Energy Consumption (EC). Hierarchical, flat, and location-based routing are the three network models. Routing functions such as delivery mechanisms and computation methods are used to operate the system [7]. Wireless Sensor Networks (WSN) link the virtual and physical worlds—tiny sensors coupled to one another transfer values to the Internet. Sensors are implemented in the network to monitor many physical and environmental features in a WSN. This network comprises many nodes, each of which has a particular function. Using Internet Protocol (IP)-based WSN technologies, the WSN can be connected to the internet [8].

WSNs are made up of nodes that can be used in several applications, including health care, agricultural, and military applications [9]. WSNs are used to monitor many applications over the internet. It permits physical objects linked together through nodes, activators, and connections to collect data. Several academics proposed WSN systems built for real-world applications based on different requirements. It describes how WSN is a backbone for IoT by considering an example of smart cities [10]. Connecting sensors, lights, and meters are the IoT devices that smart cities employ to collect and evaluate data. After obtaining this data, smart cities can use it to enhance and improve their public services, infrastructure, and amenities. An intelligent, secure environment is within approach with the support of these intelligent things.

A division of the IoT that will end up in improved transport networks is called the Internet of Vehicles (IoV). In IoV, there are three communication possibilities: Vehicles-to-Vehicles (V2V), Vehicles-To-Roadsides (V2R), and Roadsides-To-Roadsides (R2R). VANETs are utilized in several applications, including vehicle speed monitoring, traffic conjunction avoidance, optimal route finding, and communication between outside and inside cars [11]. As a result, WSNs are a collection of specialized objects that provide sensing services to IoT devices. When it involves detecting, monitoring, and tracking different environmental tasks, WSNs are essential networks. The rapid development of communication and networking systems has resulted in a demand for the planning, building, and executing of multiple intelligent (unmanned) service networks [12].

In optimizing and designing adaptive switching and routing protocols for the IoT, probabilistic models play a prominent role. By examining patterns of network traffic and device connections, these models permit effective routing decisions based on dynamic environments. A case study could emphasise employing a protocol in a smart city development, where IoT devices connect for numerous applications like environmental monitoring, traffic management, and public services. Through continuous adjustment and monitoring of routing approaches utilizing probabilistic models, the network can adapt to altering conditions, optimize resource utilization, and ensure reliable communication within the IoT ecosystem.

1.1. Contribution of the research

(a) Using a successful routing design with RPL, this study intends to improve link accuracy, EC, and QoS in WSNs.
(b) To boost the link accuracy in WSN by developing the dynamic and EC effective parents' decision in RPL.
(c) To determine efficient software-defined networking-based WSN using Machine Learning (ML) to enhance QoS in WSN.

Select the optimum parent according to the parent's predicted energy to try out the RSS system.

The remainder of the article is organized as follows: Section 1: the overview of IoT, WSN, and routing protocols are detailed in the Introduction; Section 2: the related work, comprehensive analysis, research gap, motivation, and objective of the research are given in Related Works, Section 3: the proposed Adaptive switching and routing protocol based on probabilistic model is given in Proposed methodology, Section 4: the simulation analysis with illustration is shown in Result and Discussion, and Section 5: the article is concluded with recommendation for future is presented in Conclusion.

2. Related Works

For the routing process, the Routing Protocol for Low-Power and Lossy Networks (RPL) protocol was enhanced as a well-known resolution in its Machine-to-machine (M2M) connection because of RPL's precise flexibility. The RPL routing's qualifications have been prepared to meet several requirements of dissimilar Low Power and Lossy Networks (LLN) applications. The principal uniqueness of the RPL protocol can be estimated in this research work, where the crucial purpose is to provide the reader with complete information about the RPL protocol. In RPL, numerous problems need to be located for the ever-present utilization of applications of the LLNs in the physical surroundings [13].

The optimal route amongst two Sensor Nodes (SN) in the LLNs cannot be provided by default since the buffer's size and memory are limited. Consequently, path discovery will be required to sustain the Point-to-Point (P2P) communication. In a reactive routing protocol, the optimal route has been discovered by applying the P2P-RPL. This has led to a considerable overhead in large and dense networks [14]. In the network, all the SNs in WSN need to cooperate in structuring the provisional Destination Oriented Directed Acyclic Graph (DODAG) during the pathfinding [15]. However, this is directed to enhance EC. In LLNs, it is used to discover the optimal route that considers EC. In the protocol structure, the reliability must be measured [16].

For universal data traffic, LLN purposes required fixed routing supportiveness. The P2P, Point-to-Multipoint (P2MP), and Multipoint-to-Point (MP2P) with heterogeneous facilities are obtainable for the SN in WSN. The RPL has constructed the DODAGs in behaviour that can be used in the routing to optimize MP2P data traffic. However, the P2P and P2MP's broadcasted information must move on to the pre-designed DAG method. Consequently, RPL has not provided the effectual routing, and the message delivery goes through from high EED and lossy wireless links. The P2P communication required by frequent LLN appliances must be maintained by a mediator agent in the process of RPL [17].

The DODAG root, in the manner of a non-storing procedure, has denoted an intermediate SN. As a result, it can be restricted access with a high range of data traffic. In the DODAG method, the P2MP data traffic is sustained by utilizing downward paths. Besides, the downward routes may not be optimal since the wireless channels contain asymmetric behaviour. Thus, this type of asymmetric character of wireless links has affected the routing algorithms' performance in LLNs. While they have been utilized for asymmetric links, the protocols do not carry the asymmetry of links that can provide fail-in real-time networks [18].

The traffic conjunction has taken place in LLN for both the connection and levels of an SN in a WSN. In the case of more traffic, the inadequate node's buffer size can be directed to the overflow process [19]. Besides, the traffic conjunction will be directed to reducing channel quality and PLR extension. Thus, this will lead to more EEDs, struggle on the channel, and packet drop, which will also be enhanced. Currently, many techniques are being developed to overcome traffic

conjunction problems, such as [20].

For LLNs, two significant objectives must be used to enhance routing protocols: energy-saving and reliable data or message delivery. However, multicast routing is maintained by the RPL rouging, and it can select one SN as a superior parent exploiting definite object tasks. In the lossy wireless channels and dynamic changing topology, this single route routing cannot be sufficiently measured [21]. In wireless channels, packets' data, packet rebroadcasting, and PLR have been enhanced during this unreliability process. This will be directed to a high level of EC and an extended time of channel possession.

The challenges of RPL have been discussed under severe conditions, including the discovery of energy-efficient routes, effectual routine maintenance for general traffic patterns with inadequate memory, Jamming finding and controlling, Energy-efficient and Reliable routing in LLNs, High percentage of throughput with a low duty cycling of lower layer, mobility support, issues in security method, complexity of RPL, under specification of RPL protocol, trickle junction and loops.

3. Research gap

The research gap lies in addressing the limitations and challenges of the RPL protocol in LLN. Despite its flexibility, RPL faces suboptimal routing, energy inefficiency, and susceptibility to jamming. Additionally, there's a need for more efficient multicast routing and consideration of dynamic network conditions. Addressing these gaps could enhance routing reliability and energy efficiency in LLN.

3.1. Motivation

Modern routing algorithms employed by LLN present multiple errors and issues, which is why this research is needed. After a thorough review, it has been concluded that these techniques require significant changes in order to provide accurate and effective routing. So, to enhance WSN-QoS, EC, and total accuracy, novel and better routing methods are essential. Route and data distribution efficiency is the highest priority today due to the increasing importance of IoT technologies.

New methods for cluster tasks, Cluster Head (CH) selection, traffic simulation, and routing efficiency are the main objectives of this study, which attempts to address this vital requirement. Using cutting-edge optimization techniques, including Fuzzy Assisted Cuckoo Search Optimization (FACSO), Gaussian Process Regression (GPR), and Stochastic Optimization Algorithms (SOA), we can improve cluster tasks for optimal NL or reduce EC by decreasing the impact on unpredictability in node features and the state of the network. Node features such as battery level, processing capability, and communication range influence task allocation and routing decisions. Network state variables like traffic load and congestion levels guide efficient data distribution and minimize energy wastage. Investigation significantly improves End-to-End Delay (EED), Network Throughput (NT), and EC in WSNs by implementing new routing optimization methods like FACSO, which update real-time routing decisions based on WSN circumstances. The intention is to address the challenge of practical and unpredictable connectivity in IoT-enabled settings by addressing the drawbacks of present-day routing technologies.

4. Proposed model-adaptive switching and routing protocol based on probabilistic models (ASRP-PM)

The proposed technique, ASRP-PM, is accomplished via a clustering scheme that devices to assign EC between the clusters at a stable time, thus developing NL. The primary intent of ASRP-PM is to minimize the EC across the sensor device, reduce the EED, and reduce message overhead for the admittance of real-time data. The NL of WSN is exploited.

4.1. Network formulation

The dynamic WSN is expressed via an undirected graph, denoted by the G=(L, SN). In this graph, the accumulation of links among the SN is denoted as *'L,'* and the set of SN is denoted as SN. The link between the SN '*y*' and '*x*' in graph G, where the nodes rely on the communication range.

The core concept of WSN is defined as follows.

(a) Nodes residing in the network pose unique node identity (ID), and before the incidence of fusion, there was no awareness about the network other than its ID. Entire node information will be removed once it leaves the network region, and the node ID will be for further transmission.
(b) In the WSN, the Base Station (BS) is available in the preceding link, and the setup of BS has already been attained.
(c) The distinct node can replicate the response or broadcast, which accomplishes a local estimation and is fixed interval and coordinated time. Every period, a single node is activated as a receiver or transmitter, and in some instances, both will act together. The data in the sink or gathering gaps are designated as round.
(d) Collusion identification is not available in the network.
(e) The nodes in the adjacent position can communicate, and the information among the nodes is symmetric. It depicts a specific node, and '*y*' permits the data from the '*x*' node. The node '*x*' also allows the message from '*y*'.

Nodes in the graph are segregated into several groups that are stated as clusters, and these clusters accomplish G's sub-graph. In the formulated cluster, a distinct node that acts as a center for the sub-graph, the CH, exists and is linked with all the nodes. The cluster members generate a form of network topology.

4.2. Cluster task

In WSN, the CH selection technique is attained by the fitness value utilized to elect CHs from the SN. The Fitness Function (FF) encompasses two significant measures: the usage of energy by the CH and nodes in the network. Secondly, total EC combines the information at the CH point and propagates the message to the BS. The EC between the CHs and SN is valued by EQU (1) to EQU (4).

$$Eg_i = \sum_{j=1}^{j} \sum_{\forall n_{j_i \in j}} \left\{ \frac{Eg_i - w_{fitness_fun}(j_i)}{Eg_{maxi} - Eg_{mini}} \right\} \times j_i \tag{1}$$

$$w_{fitness_{fun}}(j_i) = op_lvl(i) + WT + D_c + R_m \tag{2}$$

$$WT = WT_{maxi} \left[1 - \frac{Eg_{residual_energy}}{Eg_{initial_energy}} \right] \times Avg\left(v_j^i\right) \tag{3}$$

$$D_c = \left(\frac{n_i}{distance(i,j)} < t_{range} \right) \tag{4}$$

where the level of operator is indicated as *op_lvl*, the weight of the node is indicated as WT, and the relative mobility is indicated as 'R_m' that occurs among the node and CH, and the degree of connectivity is indicated as 'D_c' where all the nodes in the network form the cluster.

Stochastic optimization-based routing utilizes stochastic optimization techniques to challenge the uncertainties and variabilities present in network environments. Initially, the problem of routing is identified by employing this approach, implementing the advantage of unpredictability like link quality, traffic flow, and failures of nodes. Then, adopting a model of stochastic challenges, targets like optimizing NT or minimizing EED are determined. EQU (5) defines the Objective Function (OF) for maximizing the routing efficiency, F(x).

$$f(x) = \sum_{(i,j)\in E} \mu_{ij} x_{ij} \tag{5}$$

where the expected EED is given as (*i, j*).

It defines the problem in the context of stochastic variables, such as the energy of nodes and traffic flows. OF aims to improve parameters such as NL or EC within stochastic constraints. Selections are driven by stochastic designs, which describe limitations. In order to effectively address the issue, SOA techniques such as Genetic Algorithms (GA) are used. To ensure adaptability, adaptive processes change CH selection following real-time data. The technique has been enhanced via testing and parameter setting. Applying this method, we may improve reliability and performance in dynamic network environments through the development of CH methods of selection that make the most of stochastic optimization. In Algorithm 1, we can observe the phases of the recommended method.

Algorithm 1 for ASRP-PM

Step 1. Function Cluster Task WSN
Step 2. For Each **SN '*i*' in WSN**
Step 3. Compute ***E_g_i* using EQU (1)**
Step 4. Compute ***w_FF (j, i)* using EQU (2)**
Step 5. Compute ***WT* using EQU (3)**
Step 6. Compute **D_c using EQU (4)**
Step 7. For Each **SN '*i*' in WSN:**
Step 8. For Each **neighbouring node '*j*':**
Step 9. If ***E_g_i-w_FF (j,i)*** $\geq$ ***E_g_Max_i - E_g_Min_i***
Step 10. Assign node '*j*' to cluster of node '*i*'
Step 11. Update cluster task probabilities based on traffic models
Step 12. Apply SOA-based routing to optimize cluster tasks
Step 13. Return **optimized clustered WSN**
Step 14. End

4.3. Probabilistic traffic model using GPR

Regression and probability classification problems are solved using the general supervised ML technique, GPR. For small data sets, it accurately forecasts the uncertainty measures. By giving each function a probability value and calculating the mean of the probability distribution among the functions, it can identify the function that best fits the data out of many tasks while accurately describing the facts. The predictable regression function has a degree of confidence in this probabilistic technique.

The next value is computed using data flow, and the prior function space is stated to compute the predictive subsequent distribution. As in multivariate Linear Regression (LR), in GPR, time is also reserved, all nodes in the path, initial and final packets are measured as independent variables, and status is a dependent variable.

By giving the function $y = \boldsymbol{\theta} x + e$, the probability function $p(\theta)$ is computed. $P(\theta)$ represents the previous distribution on the parameter 'θ'. The subsequent distribution is computed using the Bayes rule given in EQU (6).

$$P\left(\frac{\theta}{y}, x\right) = \frac{P\left(\frac{y}{x}, \theta\right) p(\theta)}{p\left(\frac{y}{x}\right)} \tag{6}$$

where $P\left(\frac{\theta}{y}, x\right)$ is the subsequent distribution, $P\left(\frac{y}{x}, \theta\right)$ is the probability and prediction of $\frac{y}{x}$ being true if the given θ is true $p\left(\frac{y}{x}\right)$, and is the marginal probability.

Then, this model's GPR of prediction of the 'y' value is assumed in EQU (7).

$$p(D, y, x) = \int_{\theta} p(D, \theta) p\left(\frac{\theta}{y}, x\right) d\theta \tag{7}$$

where 'y' is the prediction label, and 'x' is the test observation.

Table 1
Fuzzy variable range.

Node Features	Fuzzy Variable	Input Range
Node Degree	Low	Less Than 1
	Medium	1 to 4
	High	Greater Than 4
Node Energy	Low	0.0–0.35
	Medium	0.36–0.65
	High	0.66–1
Node ED	Low	Less Than 1 ms
	Medium	1 ms–3 ms
	High	More than 3 ms

4.4. Fuzzy Assisted Cuckoo Search Optimization-based routing

This proposed FACSO approach is implemented in the controller node. The parameter node degree, EED, and energy are collected from the nodes and passed as the input to the fuzzy system. The node with high energy participates in the Cuckoo Search (CS) algorithm to find the optimal path from the SrN to DsN.

The FL system assumes input as node degree, energy, and EED and produces the output, whether the node is intruder or regular.

This fuzzy system works as a three-step process.

- **Fuzzification:** In this step, the complex input of node degree, energy, and EED is converted into a fuzzy set.
- **Inference:** Based on the provided rules, it applies the fuzzy input set and produces the fuzzy output.
- **Defuzzification:** In this step, the predicted output is converted to a complex value, and the node is determined as either an intrusion or a normal node.

The following are the linguistic and fuzzy input variables needed to find intrusion nodes.

- Node Degree – (Low, Medium, High)
- Node Energy Level – (Low, Medium, High)
- Node EED- (Less, Normal, More)

The input range of each fuzzy variable is given in Table 1. The linguistic variables used for the output variable node type are Rather Risk, Risk, High Risk, Low Risk, Moderate, Low Safe, High Safe, Safe, and Rather Safe.

Rule 1: If the node degree is > 1, then the node degree is Low, indicating a sparse network connectivity.

Rule 2: If the node degree is between 1 and 4, then the node degree is Medium, suggesting moderate network connectivity.

Rule 3: If the node degree is < 4, then the node degree is High, indicating dense network connectivity.

Rule 4: If the node energy is between 0.0 and 0.35, then the node energy is Low, implying low energy reserves.

Rule 5: If the node energy is between 0.36 and 0.65, then the node energy is Medium, suggesting moderate energy reserves in the node.

Rule 6: If the node energy is between 0.66 and 1, then the node energy is High, indicating ample energy reserves in the node.

Rule 7: If the node EED is less than 1 ms, then the node EED is Low, indicating minimal EED in data transmission.

Rule 8: If the node EED is between 1 and 3 ms, then the node EED is Medium, suggesting moderate EED in data transmission.

Rule 9: If the node EED is more than 3 ms, then the node EED is High, indicating significant EED in data transmission.

The CS algorithm is a meta-heuristic algorithm that uses the FF to approximate the fitness level of a node using the parameters link NL, bandwidth, and node energy. The FF is used for steadier routes, thus improving the Packet Received Rate (PRR). The FF used in this approach is the primary function, the sum of energy, NL, and bandwidth param-

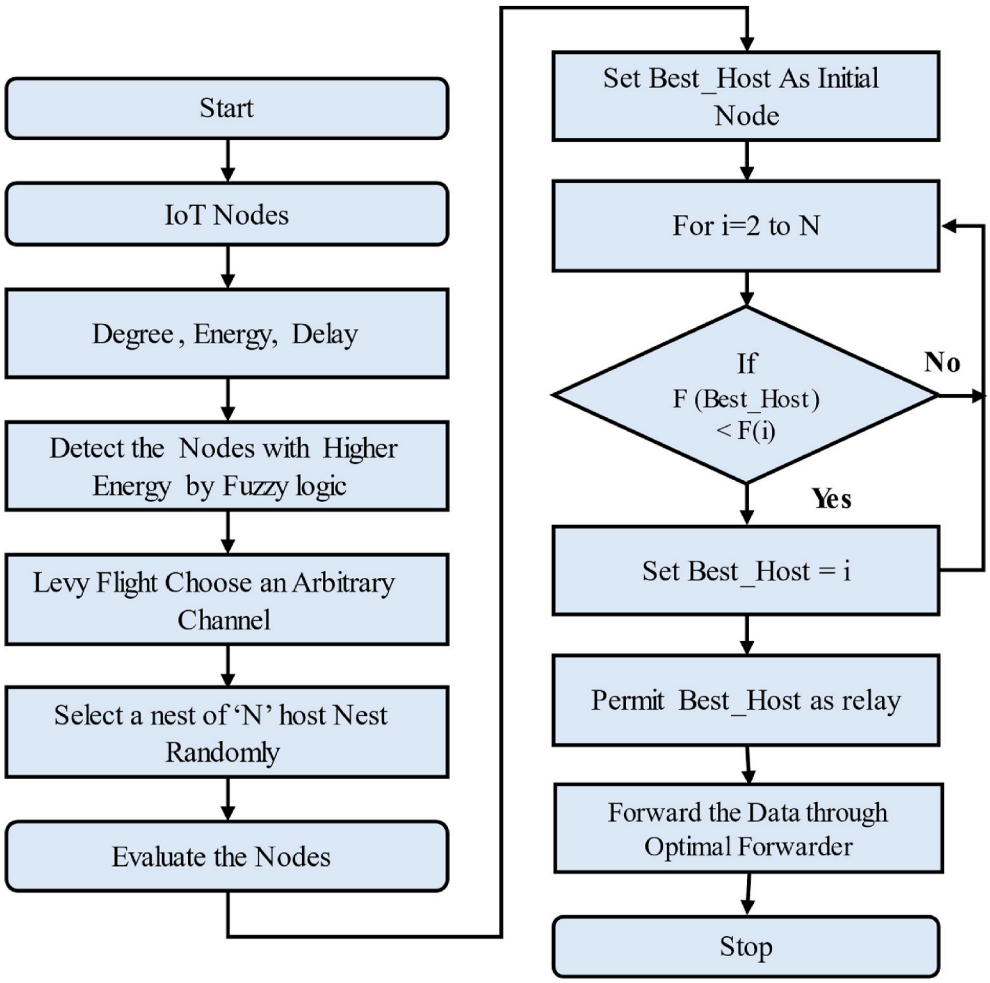

Fig. 1. Flowchart of FACSO

eters calculated. The calculation of FF is given in EQU (8).

$$f(x_i) = \frac{\sum (Energy + bandwidth + lifetime)}{Distance(x_i - x_{i-1})} \tag{8}$$

The parameters link NL, energy, and bandwidth used in the FF calculation are computed as follows:

The time required for two nodes to communicate over a route is known as the Link Lifetime (LL). In order to identify a significant secure path, a factor analysis is performed. Data Packet Loss Rate (PLR) can be decreased with the support of this secure route direction. It is well-defined as the variance between the distance Δ_d over time Δ_t, EQU (9).

$$LL = 1 - \frac{\Delta d}{\Delta t} \tag{9}$$

The lower probability of link failure and higher communication effectiveness of data packets define efficient LL connecting two nodes. Bandwidth analysis is used to avoid data PLR in WSN. The study also resulted in minimizing traffic. Consider that CB refers to bandwidth capacity and SQ to backlog size. EQU (10) provides the Bandwidth Availability (BA).

$$BA = 1 - \frac{SQ}{CB} \tag{10}$$

Each node in the network was first provided with an appropriate level of energy—the sending and receiving of packets needed a particular level of energy from the nodes. EQU (11) provides the process for computing the node's residual energy level.

$$Energy = Initial\ Energy - Utilized\ Energy \tag{11}$$

In order to identify the optimal relay node, all three of these variables are employed to determine the CS fitness. Fig. 1 demonstrates the flow chart of the FACSO approach. Each network node's parameters, such as degree, EED, and energy, are initially inputted into the FL system to identify high-energy nodes. A node with high energy will participate in the CS algorithm. Levy flight chooses a random channel and detects the neighbourhoods of the current node '*i*'.

Each neighbourhood node is evaluated using the FF. The best one is selected as the next hop node among the neighbours. Similarly, from the new hop, the process repeats to choose the next hop in the path. This process continues until it reaches the final destination node.

The procedure of the FACSO is assumed in the Algorithm 2.

Algorithm 2. Procedure of FACSO

Step 1. Start FACSO
Step 2. Get Node Degree, Energy, EED
Step 3. Apply Fuzzy Logic System
Step 4. If (Node has the highest EED, highest energy, and lesser node connectivity)
Step 5. Node with higher energy
Step 6. Else
Step 7. Normal node
Step 8. End Fuzzy logic system
Step 9. Launches CS
Step 10. Measure its fitness Fl
Step 11. If ($F_k < F_l$)
Step 12. Update l as a new solution

(*continued on next page*)

Table 2
Comparison of NT (bytes).

No of Node	RPL	LLN	FACSO
100	2904	2504	8404
200	4358	4138	16,495
300	5451	5101	21,264
400	5997	5787	24,659
500	6895	6295	27,177

Table 3
Comparison of EED (*ms*).

No of Node	RPL	LLN	FACSO
100	0.199922	0.181922	0.141717
200	0.691281	0.491201	0.294001
300	0.965479	0.800479	0.446285
400	1.56976	1.10976	0.598570
500	1.83556	1.40357	0.743240

Table 4
Comparison of PRR (count).

No of Node	RPL	LLN	FACSO
100	153	150	151
200	281	284	264
300	289	300	330
400	313	346	378
500	387	400	413

(*continued*)

Algorithm 2. Procedure of FACSO

Step 13. Continue the best solutions
Step 14. End CS
Step 15. Transmit the Data through the best solution
Step 16. Stop FACSO
Step 17. End If
Step 18. End

5. Result and Discussion

The FACSO improved routing efficiency and was implemented using the NS-2 simulator with 500 nodes. Any source that requests to transmit the data requests the controller node for the entire route to the destination. The controller uses the CS algorithm to find the optimal path from the source to the destination. The simulation setup encompasses different parameters defining the features and conditions of the simulated WSN environment. It employs a wireless channel with a 2-way ground radio-propagation model, Wireless PHY network interface type, and 802_11 MAC type for communication. Packet queuing is managed by a PriQueue, with LL as the link layer type and Omni_Antenna as the antenna model.

The interface queue is configured to accommodate a maximum of 50 packets. The simulation involves 500 mobile nodes employing the 6LoWPAN routing protocol to facilitate communication. The topography of the simulated network is defined with dimensions of 900 by 614 units. The simulation duration is set to 20.0 units of time. These parameters collectively establish the framework for conducting simulations to evaluate the performance and behaviour of wireless networks under specified conditions and scenarios. The measures throughput, PRR, EC, and EED are used to show the performance of the IoT end nodes using the FACSO for optimal route finding against RPL and LLN (see Table 4).

NT is a rate that measures the number of data sent within a particular time in the network's bandwidth. The total amount of time used for transmitting a packet from the initial node to its destination of choice is indicated by time EED. PRR is determined as the ratio of the total packets reached through the target node from the complete packets sent. The PRR is the ratio of the number of packets delivered to the target node to the total number of packets sent. The performance of the algorithms are given in Table 2 to Table 5. The performance is measured using EQU (12) to EQU (15) (see Table 3).

Table 5
Comparison of EC (*J*).

No of Node	RPL	LLN	FACSO
100	0.797	0.767	0.456
200	0.834	0.782	0.493
300	0.867	0.845	0.534
400	0.898	0.856	0.552
500	0.899	0.887	0.576

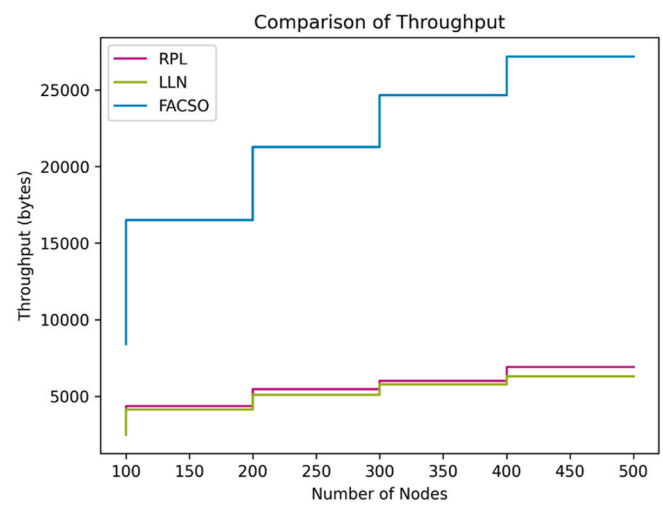

Fig. 2. Comparison of NT

$$NT = \frac{Total\ No.\ of\ Bytes\ transmitted}{Time\ Taken} \tag{12}$$

$$EED = Time\ at\ packet\ received - Time\ at\ packet\ transmitted \tag{13}$$

$$PRR = \frac{No.\ of\ Packet\ Reached}{No.\ of\ Packet\ Sent} \tag{14}$$

$$E_{tx} = P_{tx} \times T_{tx} \tag{15}$$

where the EC during data transmission is given as E_{tx}, duration is T_{tx} and energy is P_{tx}.

6. Discussion

The comparative analysis of the routing protocols, RPL, LLN, and FACSO, across several performance metrics reveals distinct performance differences. FACSO consistently outperforms RPL and LLN in terms of NT, with significantly higher NT values recorded across all numbers of nodes: 8404, 16,495, 21,264, 24,659, and 27,177 bytes for 100, 200, 300, 400, and 500 nodes, respectively, compared to RPL and LLN. However, FACSO exhibits higher EED values compared to RPL and LLN, with delay values ranging from 0.141717 ms to 0.743240 ms for FACSO across different numbers of nodes, while RPL records the lowest EED values ranging from 0.199922 to 1.83556 ms.

Additionally, FACSO demonstrates lower EC than RPL and LLN, with energy consumption values of 0.456–0.576 *J* for FACSO, whereas RPL and LLN show relatively higher EC, ranging from 0.767 to 0.899 *J*.

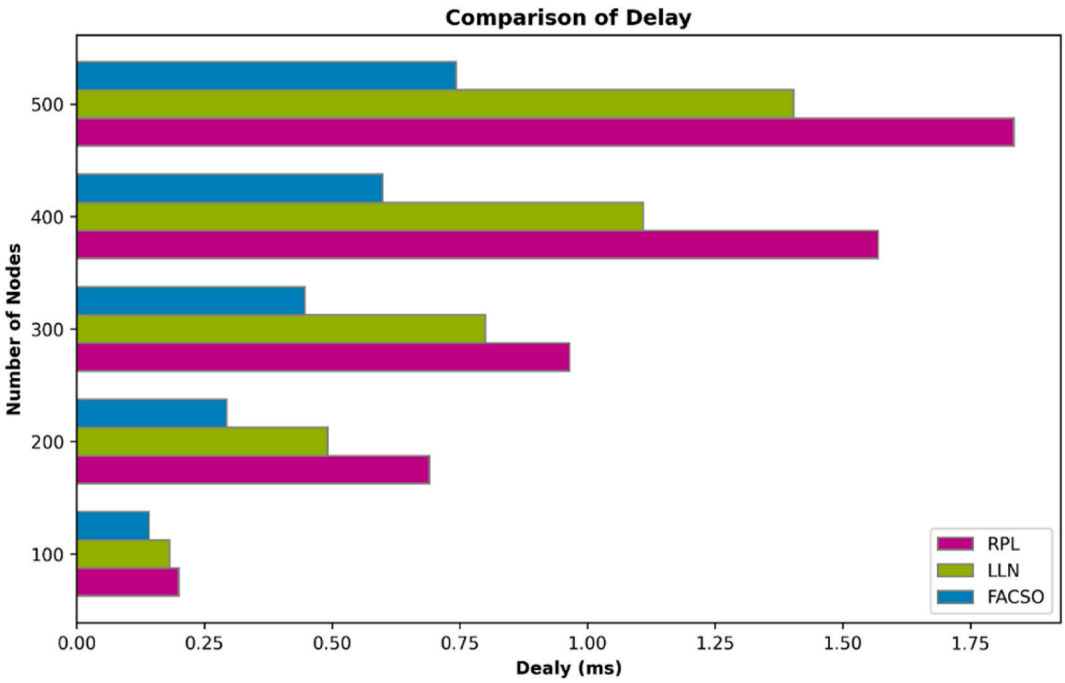

Fig. 3. Comparison of EED

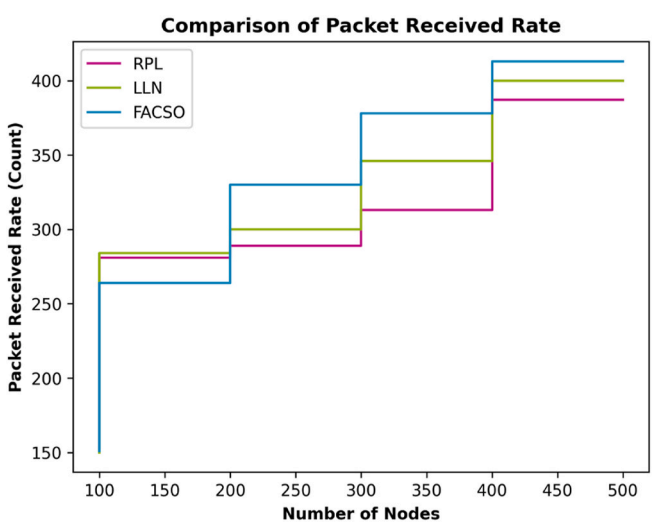

Fig. 4. Comparison of PRR

Despite these differences, all three protocols exhibit similar trends in PRR, with minor variations observed across different numbers of nodes. These numerical comparisons underscore FACSO's potential as a more efficient routing protocol, particularly in scenarios prioritizing high NT and energy efficiency.

The NT, EED, PRR, and EC analysis reveals distinct performance features among the RPL, LLN, and FACSO routing protocols. In contrast to LLN and FACSO, RPL consistently shows lower NT across numerous nodes, with FACSO consistently achieving the greatest NT. This structure is depicted visually in Fig. 2, where the highest NT is identified in FACSO, followed by LLN and RPL. Compared to LLN and FACSO, RPL has the most rapid data transmission due to its minimal EED ratios. However, the highest EED ratios can be detected in FACSO, which might suggest that data transmission EED is not practical.

The variances are illustrated visually in Fig. 3, where the lowest delay can be seen with RPL next to LLN and FACSO. As demonstrated by Fig. 3, PRR tends to be constant within RPL, LLN, and FACSO, even when the total amount of nodes changes significantly. Yet FACSO consistently displays lower EC than RPL and LLN, indicating it could be the better EC approach.

Fig. 4 illustrates this finding, suggesting that FACSO frequently shows lower EC than RPL and LLN. In general, when comparing NT, EED, and EC, FACSO performs more effectively than RPL and LLN, suggesting that it could be an improved routing protocol. Testing the results and investigating the hidden variables resulting in performance variations involves more investigation and testing (see Fig. 5).

Once validated through simulations, the adaptive protocols are deployed in a controlled environment within the company's facilities. Real-time monitoring tools track key performance metrics such as PDR, EED, and NT.

7. Conclusion and future work

The Internet of Things has brought given about transformative alterations across diverse domains, namely Intelligent Transportation Systems, smart cities, wearable devices, and agriculture. This research work focuses on improving energy efficiency in Internet of Things devices by enhancing traffic prediction, data flow, and decision-making system of cluster heads. With the assistance of the innovative applications of Fuzzy Assisted Cuckoo Search Optimization, Gaussian Process Regression, and Stochastic Optimization Algorithm, the research prolongs network lifetime while minimizing energy consumption. The real-time decision-making system for routing, utilizing Fuzzy Assisted Cuckoo Search Optimization, enhances network throughput, end-to-end delay, and energy efficiency but also improves the consistency of wireless sensor networks. This novel technique offers promising prospects for optimizing Internet of Things device performance and network consistency in diverse applications.

Future innovations may include using artificial intelligence and optimized performance methods motivated by nature.

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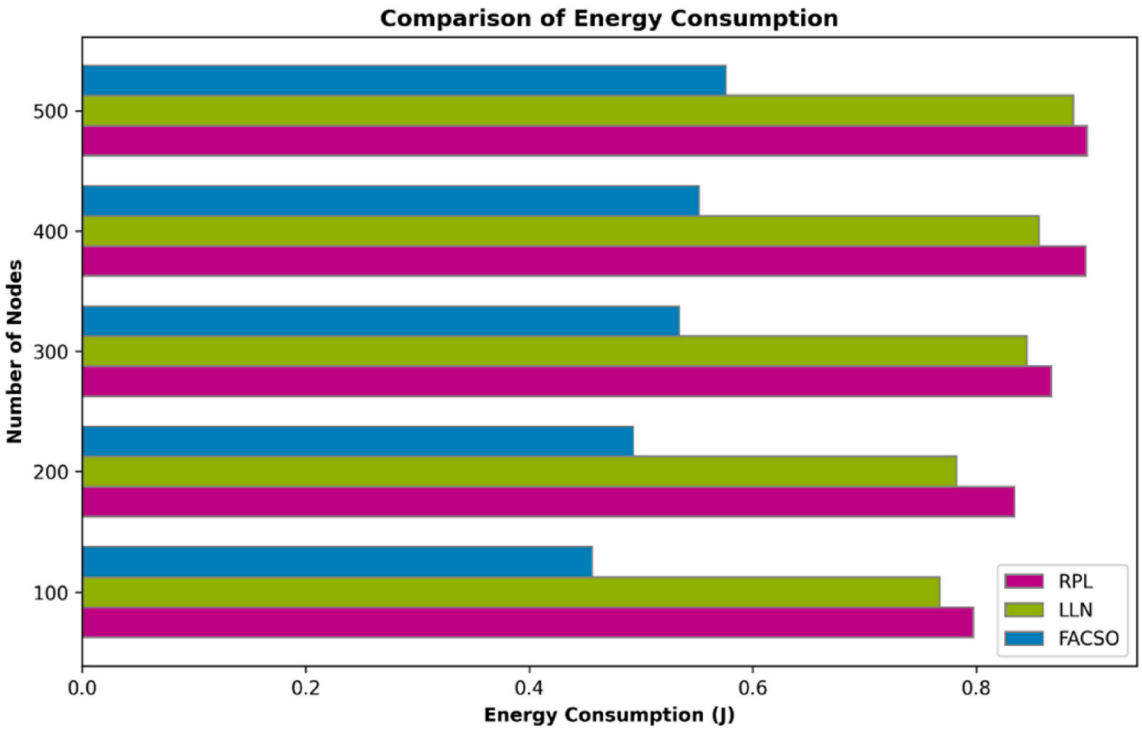

Fig. 5. Comparison of EC

Data Availability

Not Applicable.

CRediT authorship contribution statement

Yi Yang: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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